

Exercise #8.5

Q#1: Show that the straight line $\frac{x+3}{2} = \frac{y-4}{-7} = \frac{z}{3}$ is parallel to the plane $4x + 2y + 2z = 9$.

Solution:

Given equation of line and plane

$$\frac{x+3}{2} = \frac{y-4}{-7} = \frac{z}{3} \quad \text{--- (L)}$$

$$\& \quad 4x + 2y + 2z = 9 \quad \text{--- (P)}$$

Direction ratios of line L are $a_1 = 2, \quad b_1 = -7, \quad c_1 = 3$

Now direction ratios of normal vector of plane P are $a_2 = 4, \quad b_2 = 2, \quad c_2 = 2$

We have to prove $P \parallel L$

If $P \parallel L$ then normal vector of plane P is perpendicular to the line L. we have

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

$$(2)(4) + (-7)(2) + (3)(2) = 0$$

$$8 - 14 + 6 = 0$$

$$0 = 0$$

Hence proved that the line L and plane P are parallel.

Q#2: Show that the straight line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ is perpendicular to the plane $4x + 8y + 12z + 19 = 0$.

Solution:

Given equation of line and plane are

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{3} \quad \text{--- (L)}$$

$$4x + 8y + 12z + 19 = 0 \quad \text{--- (P)}$$

Direction ratios of line L are $a_1 = 1, b_1 = 2, c_1 = 3$

and direction ratios of normal vector of plane P are $a_2 = 4, b_2 = 8, c_2 = 12$

We have to prove $P \perp L$

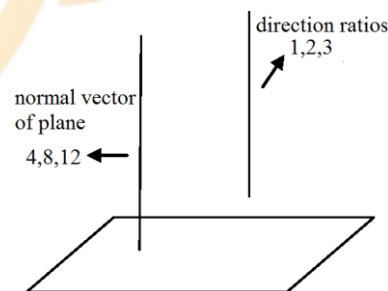
If $P \perp L$ then normal vector of plane P is parallel to the line L. we have

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{1}{4} = \frac{2}{8} = \frac{3}{12}$$

$$\Rightarrow \frac{1}{4} = \frac{1}{4} = \frac{1}{4}$$

Hence proved that the line L and plane P are perpendicular.



Q#3: Find the condition that the straight line $x = mz + a, y = nz + b$ may lie the plane $Ax + By + Cz + D = 0$.

Solution: Given equations of line and plane are

$$x = mz + a, \quad y = nz + b$$

$$\& \quad Ax + By + Cz + D = 0 \quad \text{--- (P)}$$

From given equation of line is

$$x = mz + a \Rightarrow z = \frac{x-a}{m} \quad \dots \dots (1)$$

$$y = nz + b \Rightarrow z = \frac{y-b}{n} \quad \dots \dots (2)$$

$$\Rightarrow \frac{x-a}{m} = \frac{y-b}{n} = \frac{z}{1}$$

Let $\frac{x-a}{m} = \frac{y-b}{n} = \frac{z}{1} = t \quad (\text{say})$

Then $\begin{cases} x = a + mt \\ y = b + nt \\ z = t \end{cases}$

Given that line L lies on the plane P then each point of line lies on plane. Hence point $(a + mt, b + nt, t)$ lies on plane. So equation (P) becomes

$$\Rightarrow A(a + mt) + B(b + nt) + Ct + D = 0$$

$$\Rightarrow Aa + Amt + Bb + Bnt + Ct + D = 0$$

$$\Rightarrow [Aa + Bb + D] + t[Am + Bn + C] = 0$$

This eq. must be satisfied for every value of

$$\Rightarrow Aa + Bb + D = 0 \quad \& \quad Am + Bn + C = 0$$

This is the required condition for which the given line lies on the given plane.

Q#4: Determine the point, if any, common to the straight line $x = 1 + t, y = t, z = -1 + t$ and the Plane $x + y + z = 3$.

Solution:

Given equation of straight line and plane are

$$\left. \begin{aligned} x &= 1 + t \\ y &= t \\ z &= -1 + t \end{aligned} \right\} \quad \text{--- (L)}$$

$$\& \quad x + y + z = 3 \quad \text{--- (P)}$$

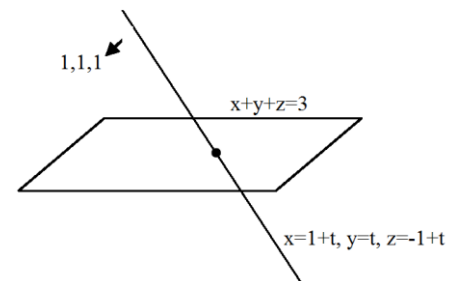
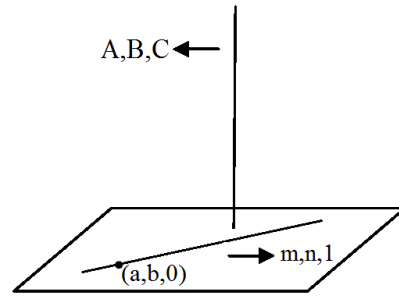
Let a point $(1 + t, t, -1 + t)$ is common point of line L and plane P.

This point satisfies the equation of the plane then

$$\Rightarrow 1 + t + t + (-1 + t) = 3$$

$$\Rightarrow 1 + t + t - 1 + t = 3 \Rightarrow 3t = 3 \Rightarrow t = 1$$

Hence the common point of the line L and plane P is $(1 + t, t, -1 + t) = (1 + 1, 1, -1 + 1) = (2, 1, 0)$.



Q#5: Find an equation of the plane through the point (x_1, y_1, z_1) and through the straight line

$$\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-c}{n}$$

Solution: Given equation of straight line

$$\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-c}{n}$$

Let A, B & C are the direction ratios of normal vector of the plane Then equation of the plane through given line will be written as

$$A(x-a) + B(y-b) + C(z-c) = 0 \quad \text{---(1)}$$

$$Al + Bm + Cn = 0 \quad \text{---(2)}$$

Required equation of the plane passes through the point (x_1, y_1, z_1) . then eq. (1) will become

$$A(x_1 - a) + B(y_1 - b) + C(z_1 - c) = 0 \quad \text{---(3)}$$

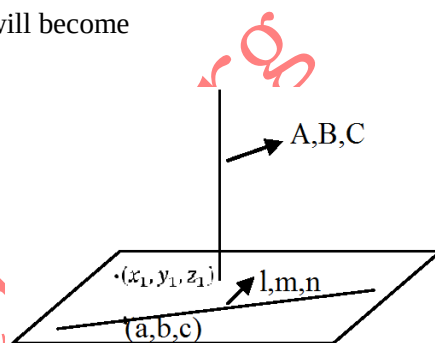
Eliminating A, B & C from eq. (1), (2) & (3) we have

$$\begin{vmatrix} x-a & y-b & z-c \\ l & m & n \\ x_1-a & y_1-b & z_1-c \end{vmatrix} = 0$$

or
$$\sum (x-a) \begin{vmatrix} m & n \\ y_1-b & z_1-c \end{vmatrix} = 0$$

$$\sum (x-a) [m(z_1 - c) - n(y_1 - b)] = 0$$

This is the required equation of plane.



Q#6: Find an equation of the plane passing through straight line $x + 2z = 4, y - z = 8$ and parallel to the straight line

$$\frac{x-3}{2} = \frac{y+4}{3} = \frac{z-7}{4}$$

Solution: Given equation of straight line

$$x + 2z = 4, y - z = 8 \quad \text{--- } L_1$$

$$\frac{x-3}{2} = \frac{y+4}{3} = \frac{z-7}{4} \quad \text{--- } L_2$$

Consider line L_1 $x + 2z = 4$, $y - z = 8$

$$\Rightarrow 2z = 4 - x, \quad z = y - 8$$

$$\Rightarrow z = \frac{x-4}{-2} \quad \text{--- (1)}, \quad z = \frac{y-8}{1} \quad \text{--- (2)}$$

Equating (1) & (2)

$$\frac{x-4}{-2} = \frac{y-8}{1} = \frac{z}{1} \quad (\text{symmetric form})$$

Let A, B & C are direction ratios of normal vector of the plane

The equation will be written as

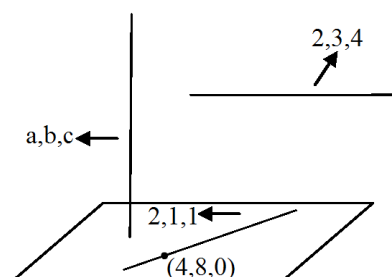
$$A(x-4) + B(y-8) + C(z-0) = 0 \quad \text{---(1)}$$

Where

$$A(-2) + B(1) + C(1) = 0 \quad \text{---(2)}$$

As required equation of the plane is parallel to the line L_2

$$A(2) + B(3) + C(4) = 0 \quad \text{---(3)}$$



Eliminating A,B & C from eq.(1),(2) & (3) we have

$$\begin{vmatrix} x-4 & y-8 & z \\ -2 & 1 & 1 \\ 2 & 3 & 4 \end{vmatrix} = 0$$

$$(x-4) \begin{vmatrix} 1 & 1 \\ 3 & 4 \end{vmatrix} - (y-8) \begin{vmatrix} -2 & 1 \\ 2 & 4 \end{vmatrix} + z \begin{vmatrix} -2 & 1 \\ 2 & 3 \end{vmatrix} = 0$$

$$\Rightarrow (x-4)(1) - (y-8)(-10) + z(-8) = 0$$

$$\Rightarrow x - 4 - (y-8)(-10) - 8z = 0$$

$$\Rightarrow x - 4 + 10y - 80 - 8z = 0$$

$$\Rightarrow x + 10y - 8z - 84 = 0 \quad \text{is required equation of the plane.}$$

Q#7: Find an equation of the plane passing through the point (α, β, γ) and parallel to each of the straight lines

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1} \quad \text{and} \quad \frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2}$$

Solution:

Let equation of the plane passes through the point (α, β, γ) with direction ratios A,B & C

$$(x - \alpha) + B(y - \beta) + C(z - \gamma) = 0 \quad \text{--- (1)}$$

Equation of the plane is parallel to the given lines

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1} \quad \text{--- } L_1$$

$$\frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2} \quad \text{--- } L_2$$

We know that normal vectors of

the plane is perpendicular to lines L_1 & L_2

Therefore,

$$Al_1 + Bm_1 + Cn_1 = 0 \quad \text{--- (2)}$$

$$Al_2 + Bm_2 + Cn_2 = 0 \quad \text{--- (3)}$$

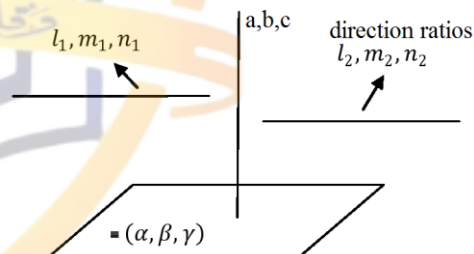
Now eliminating A,B & C from eq.(1),(2) & (3) we have

$$\begin{vmatrix} x-\alpha & y-\beta & z-\gamma \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$$

$$\text{or} \quad \sum (x-\alpha) \begin{vmatrix} m_1 & n_1 \\ m_2 & n_2 \end{vmatrix} = 0$$

$$\Rightarrow \sum (x-\alpha) (m_1 n_2 - m_2 n_1) = 0$$

This is the required equation of plane.



Q#8: Find an equation of the plane through the straight line $ax + by + cz + d = 0 = a'x + b'y + c'z + d'$ and parallel to the straight line $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$.

Solution: Given equations of the straight lines are

$$ax + by + cz + d = 0 = a'x + b'y + c'z + d' \quad \text{---} L_1$$

$$\& \quad \frac{x}{l} = \frac{y}{m} = \frac{z}{n} \quad \text{---} L_2$$

Let equation of the plane through the straight line $ax + by + cz + d = 0 = a'x + b'y + c'z + d'$ can be written as

$$(ax + by + cz + d) + k(a'x + b'y + c'z + d') = 0 \quad \text{---} (P)$$

$$ax + by + cz + d + a'kx + b'ky + c'kz + d'k = 0$$

$$(a + a'k)x + (b + b'k)y + (c + c'k)z + (d + d'k) = 0$$

$$\text{Here } A = (a + a'k), \quad B = (b + b'k), \quad C = (c + c'k)$$

are the direction ratios of the normal vector of the plane.

This plane is parallel to the line L_2

Then normal vector of the plane is perpendicular to the line L_2

$$\Rightarrow (a + a'k)l + (b + b'k)m + (c + c'k)n = 0$$

$$\Rightarrow al + a'kl + bm + b'km + cn + c'kn = 0$$

$$\Rightarrow a'kl + b'km + c'kn + al + bm + cn = 0$$

$$\Rightarrow k(a'l + b'm + c'n) + al + bm + cn = 0$$

$$\Rightarrow k(a'l + b'm + c'n) = -(al + bm + cn)$$

$$\Rightarrow k = \frac{-(al + bm + cn)}{(a'l + b'm + c'n)}$$

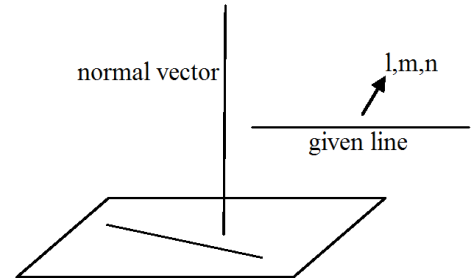
Using value of k in equation P

$$(ax + by + cz + d) - \frac{(al + bm + cn)}{(a'l + b'm + c'n)}(a'x + b'y + c'z + d') = 0$$

$$(a'l + b'm + c'n)(ax + by + cz + d) - (al + bm + cn)(a'x + b'y + c'z + d') = 0$$

$$(a'l + b'm + c'n)(ax + by + cz + d) = (al + bm + cn)(a'x + b'y + c'z + d')$$

This is the required equation of the plane.



Q#9: Prove that the straight lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ are coplanar.

Solution:

Given equations of straight lines are

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \quad \text{---} L_1$$

$$\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5} \quad \text{---} L_2$$

We have to prove that these straight lines are coplanar.

As we know that the straight lines $\frac{x-a_1}{l_1} = \frac{y-b_1}{m_1} = \frac{z-c_1}{n_1}$ & $\frac{x-a_2}{l_2} = \frac{y-b_2}{m_2} = \frac{z-c_2}{n_2}$ are coplanar if

$$\begin{vmatrix} a_2 - a_1 & b_2 - b_1 & c_2 - c_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$$

From L_1 & L_2 we have $(a_1, b_1, c_1) = (1, 2, 3)$ & $(a_2, b_2, c_2) = (2, 3, 4)$

$l_1 = 2, m_1 = 3$ & $n_1 = 4$ for line L_1

$l_2 = 3, m_2 = 4$ & $n_2 = 5$ for line L_2

Then

$$\begin{vmatrix} 2-1 & 3-2 & 4-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$$



Expanding by R_1

$$\Rightarrow 1(15 - 16) - 1(10 - 12) + 1(8 - 9) = 0$$

$$\Rightarrow 1(-1) - 1(-2) + 1(-1) = 0$$

$$\Rightarrow -1 + 2 - 1 = 0$$

$$\Rightarrow 0 = 0$$

Hence given straight lines are coplanar.

Q#10: prove that the straight lines $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ and $\frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7}$ intersect. Also find the point of intersection and the plane through them.

Solution: Given equations of straight lines

$$\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8} \text{ --- } L_1$$

$$\frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7} \text{ --- } L_2$$

Let $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8} = t$ (say) & $\frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7} = s$ (say)

Now parametric equations of given straight lines are

$$\left. \begin{aligned} x &= 1 + 2t \\ y &= -1 - 3t \\ z &= -10 + 8t \end{aligned} \right\} \text{ --- } (L_1) \quad \& \quad \left. \begin{aligned} x &= 4 + s \\ y &= -3 - 4s \\ z &= -1 + 7s \end{aligned} \right\} \text{ --- } (L_2)$$

Any point on line L_1 is $(1 + 2t, -1 - 3t, -10 + 8t)$

Any point on line L_2 is $(4 + s, -3 - 4s, -1 + 7s)$

Let a point $P(x_o, y_o, z_o)$ is a point of intersection of lines L_1 & L_2 . So this point will satisfy equations (1) & (2).

$$x_o = 1 + 2t$$

$$x_o = 4 + s$$

$$y_o = -1 - 3t$$

$$y_o = -3 - 4s$$

$$z_o = -10 + 8t$$

$$z_o = -1 + 7s$$

Equating above

$$1 + 2t = 4 + s \quad \Rightarrow \quad 2t - s = 3 \quad \text{--- (3)}$$

$$-1 - 3t = -3 - 4s \quad \Rightarrow \quad 3t - 4s = 2 \quad \text{--- (4)}$$

$$-10 + 8t = -1 + 7s \quad \Rightarrow \quad 8t - 7s = 9 \quad \text{--- (5)}$$

Now solving eq.(3) & (4)

Multiplying eq.(3) by 4 and subtracting eq.(3) & (4)

$$8t - 4s - 3t + 4s = 12 - 2 \quad \Rightarrow \quad 5t = 10 \quad \Rightarrow \quad t = 2$$

Putting value of t in eq.(3)

$$2(2) - s = 3 \quad \Rightarrow \quad 4 - s = 3 \quad \Rightarrow \quad s = 1$$

we see that these values of t & s satisfies eq.(5)

hence given lines intersect each other and their point of intersection is $(x_o, y_o, z_o) = (5, -7, 6)$

Now we have to find the equation of plane through these lines.

A plane through these lines must be contain the point of intersection of these lines.

If A, B & C are direction ratios of normal vector of the plane then equation of plane through $(5, -7, 6)$ is

$$A(x - 5) + B(y + 7) + C(z - 6) = 0 \quad \text{--- (I)}$$

As this plane contain both lines, so

$$2A - 3B + 8C = 0 \quad \text{--- (II)}$$

$$A - 4B + 7C = 0 \quad \text{--- (III)}$$

Eliminating A, B & C from eq.(I) (II) & (III) we have

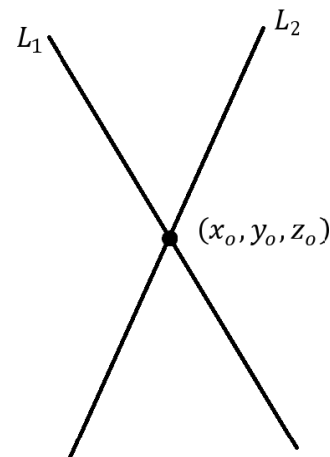
$$\begin{vmatrix} x-5 & y+7 & z-6 \\ 2 & -3 & 8 \\ 1 & -4 & 7 \end{vmatrix} = 0$$

$$\Rightarrow (x - 5)(-21 + 32) - (y + 7)(14 - 8) + (z - 6)(-8 + 3) = 0$$

$$\Rightarrow (x - 5)(11) - (y + 7)(6) + (z - 6)(-5) = 0$$

$$\Rightarrow 11x - 55 - 6y - 42 - 5z + 30 = 0$$

$$\Rightarrow 11x - 6y - 5z - 67 = 0 \quad \text{is required equation of plane.}$$



Checked by: Sir Hameed ullah (hameedmath2017 @ gmail.com)

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